

Chapter 07 Part 2: Spectroscopy

1.

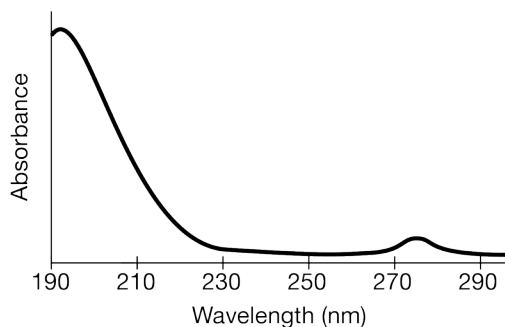


Diagram 1

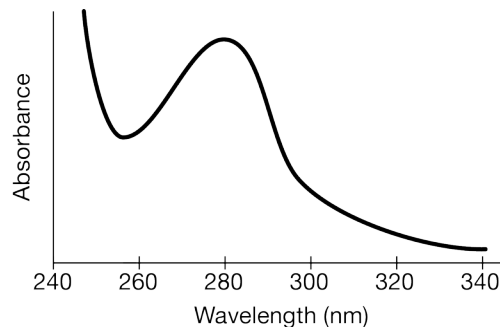


Diagram 2

The diagrams above show the ultraviolet absorption spectra for two compounds. Diagram 1 is the absorption spectrum of pure acetone, a solvent used when preparing solutions for an experiment. Diagram 2 is the absorption spectrum of the solute for which the absorbance needs to be measured to determine its concentration. When the student reads the absorbance of the solution at 280 nm, the result is too high. Which of the following is most likely responsible for the error in the measured absorbance?

- (A) The student added too little solute to the acetone before measuring its absorbance.
- (B) The student rinsed the cuvette with the solution before filling the cuvette with the solution.
- (C) The student forgot to calibrate the spectrophotometer first by using a cuvette containing only acetone.
- (D) The wavelength setting was accidentally changed from 280 nm to 300 nm before the student made the measurement.

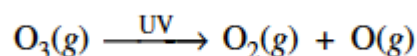
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2. Answer the following problems about gases.

- a. The average atomic mass of naturally occurring neon is 20.18 amu. There are two common isotopes of naturally occurring neon as indicated in the table below.

Isotope	Mass (amu)
Ne-20	19.99
Ne-22	21.99

- Using the information above, calculate the percent abundance of each isotope.
 - Calculate the number of Ne-22 atoms in a 12.55 g sample of naturally occurring neon.
- b. A major line in the emission spectrum of neon corresponds to a frequency of $4.34 \times 10^{14} \text{ s}^{-1}$. Calculate the wavelength, in nanometers, of light that corresponds to this line.
- c. In the upper atmosphere, ozone molecules decompose as they absorb ultraviolet (UV) radiation, as shown by the equation below. Ozone serves to block harmful ultraviolet radiation that comes from the Sun.

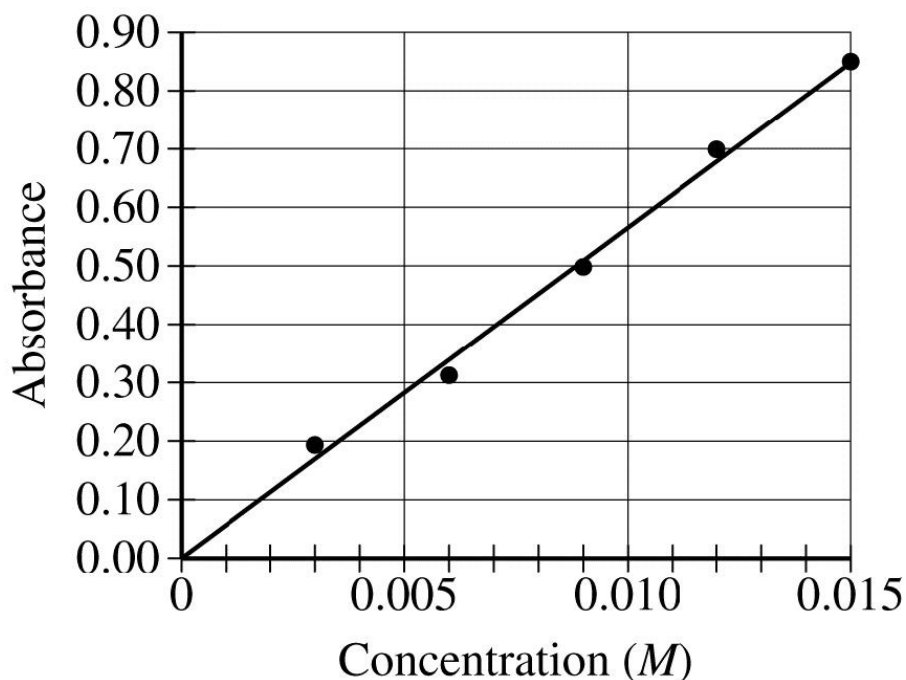


A molecule of $\text{O}_3(g)$ absorbs a photon with a frequency of $1.00 \times 10^{15} \text{ s}^{-1}$.

- How much energy, in joules, does the $\text{O}_3(g)$ molecule absorb per photon?
- The minimum energy needed to break an oxygen-oxygen bond in ozone is 387 kJ mol^{-1} . Does a photon with a frequency of $1.00 \times 10^{15} \text{ s}^{-1}$ have enough energy to break this bond? Support your answer with a calculation.

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3. To spectrophotometrically determine the mass percent of cobalt in an ore containing cobalt and some inert materials, solutions with known $[\text{Co}^{2+}]$ are prepared and the absorbance of each of the solutions is measured at the wavelength of optimum absorbance. The data are used to create a calibration plot, shown below.



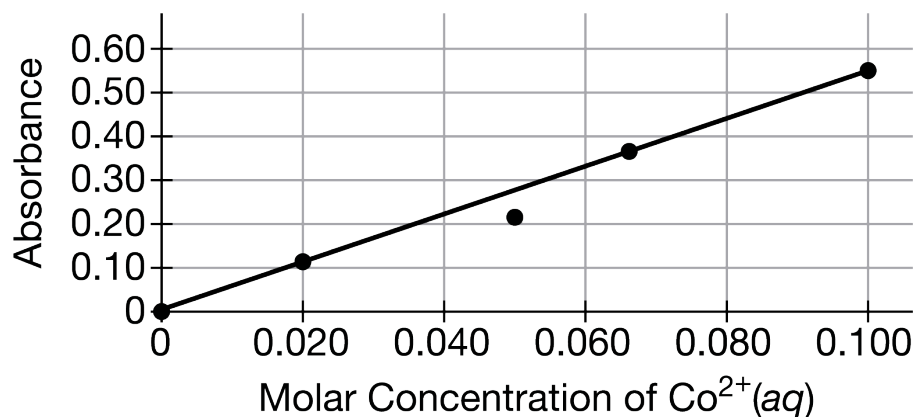
A 0.630 g sample of the ore is completely dissolved in concentrated $\text{HNO}_3(aq)$. The mixture is diluted with water to a final volume of 50.00 mL. Assume that all the cobalt in the ore sample is converted to $\text{Co}^{2+}(aq)$.

- What is the $[\text{Co}^{2+}]$ in the solution if the absorbance of a sample of the solution is 0.74?
- Calculate the number of moles of $\text{Co}^{2+}(aq)$ in the 50.00 mL solution.
- Calculate the mass percent of Co in the 0.630 g sample of the ore.

A student uses visible spectrophotometry to determine the concentration of $\text{CoCl}_2(aq)$ in a sample solution. First the student prepares a set of $\text{CoCl}_2(aq)$ solutions of known concentration. Then the student uses a spectrophotometer to determine the absorbance of each of the standard solutions at a wavelength of 510 nm and constructs a standard curve. Finally, the student determines the absorbance of the sample of unknown concentration.

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4.



The student made the standard curve above. Which of the following most likely caused the error in the point the student plotted at $0.050\text{ M Co}^{2+}(\text{aq})$?

- (A) There was distilled water in the cuvette when the student put the standard solution in it.
 - (B) There were a few drops of the $0.100\text{ M Co}^{2+}(\text{aq})$ standard solution in the cuvette when the student put the 0.050 M standard solution in it.
 - (C) The student used a cuvette with a longer path length than the cuvette used for the other standard solutions.
 - (D) The student did not run a blank between the $0.050\text{ M Co}^{2+}(\text{aq})$ solution and the one before it.
5. A wavelength of 510 nm corresponds to an approximate frequency of $6 \times 10^{14}\text{ s}^{-1}$. What is the approximate energy of one photon of this light?
- (A) $9 \times 10^{47}\text{ J}$
 - (B) $3 \times 10^{17}\text{ J}$
 - (C) $5 \times 10^{-7}\text{ J}$
 - (D) $4 \times 10^{-19}\text{ J}$
-

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6. For parts of the free response question that require calculations, clearly used the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is given the task of determining the molar concentration of a CuSO_4 solution using two different procedures, precipitation and spectrophotometry.

For the precipitation experiment, the student adds 20.0 mL of 0.200 M $\text{Ba}(\text{NO}_3)_2$ to 50.0 mL of the $\text{CuSO}_4(aq)$. The reaction goes to completion, and a white precipitate forms. The student filters the precipitate and dries it overnight. The data are given in the following table.

Mass of dry filter paper	0.764 g
Volume of $\text{CuSO}_4(aq)$	50.0 mL
Volume of 0.200 M $\text{Ba}(\text{NO}_3)_2$	20.0 mL
Mass of filter paper and dried precipitate	1.136 g

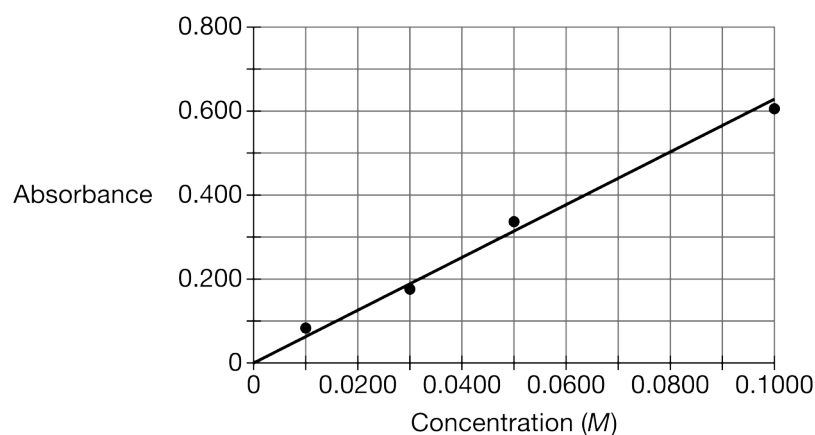
- (a) Write a balanced net ionic equation for the precipitation reaction.
- (b) Calculate the number of moles of precipitate formed.
- (c) Calculate the molarity of the original CuSO_4 solution.

For the spectrophotometry experiment, the student first makes a standard curve. The student uses a 0.1000 M solution of $\text{CuSO}_4(aq)$ to make three more solutions of known concentration (0.0500 M , 0.0300 M , and 0.0100 M) in 50.00 mL volumetric flasks.

- (d) Calculate the volume of 0.1000 M $\text{CuSO}_4(aq)$ needed to make 50.00 mL of 0.0500 M $\text{CuSO}_4(aq)$.
- (e) Briefly describe the procedure the student should follow to make 50.00 mL of 0.0500 M $\text{CuSO}_4(aq)$ using 0.1000 M $\text{CuSO}_4(aq)$, a 50.00 mL volumetric flask, and other standard laboratory equipment. Assume that all appropriate safety precautions will be taken.

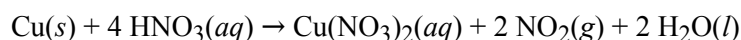
The standard curve is given below.

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(f) The absorbance of the CuSO_4 solution of unknown concentration is 0.219. Determine the molarity of the solution.

(g) A second student performs the same experiment. There are a few drops of water in the cuvette before the second student adds the $\text{CuSO}_4(aq)$ solution of unknown concentration. Will this result in a $\text{CuSO}_4(aq)$ concentration for the unknown that is greater than, less than, or equal to the concentration determined in part (f)? Justify your answer.



Each student in a class placed a 2.00 g sample of a mixture of Cu and Al in a beaker and placed the beaker in a fume hood. The students slowly poured 15.0 mL of 15.8 M $\text{HNO}_3(aq)$ into their beakers. The reaction between the copper in the mixture and the $\text{HNO}_3(aq)$ is represented by the equation above. The students observed that a brown gas was released from the beakers and that the solutions turned blue, indicating the formation of $\text{Cu}^{2+}(aq)$. The solutions were then diluted with distilled water to known volumes.

7.

$[\text{Cu}^{2+}]$	Absorbance
0.025	0.059
0.050	0.235
0.100	0.117
0.200	0.468
Unknown (from sample of mixture)	0.330

To determine the number of moles of Cu in the sample of the mixture, the students measured the absorbance of known concentrations of $\text{Cu}(\text{NO}_3)_2(aq)$ using a spectrophotometer. A cuvette filled with some of the solution produced from the sample of the mixture was also tested. The data recorded by one student are shown in the table above. On the basis of the data provided, which of the following is a possible error that the student made?

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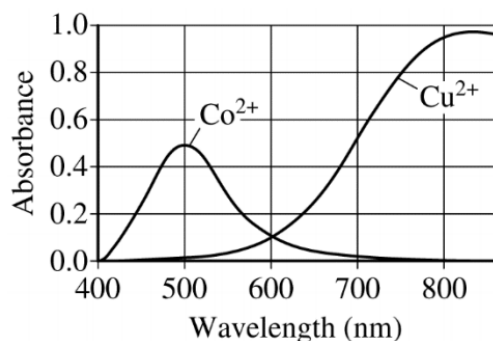
- (A) The $\text{Cu}(\text{NO}_3)_2(\text{aq})$ from the sample of the mixture was not diluted properly.
 - (B) The spectrophotometer was calibrated with tap water instead of distilled water.
 - (C) The student labeled the cuvettes incorrectly, reversing the labels on two of the solutions of known concentration.
 - (D) The spectrophotometer was originally set to an inappropriate wavelength, causing the absorbance to vary unpredictably.
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- 8. Directions:** Examples and equations may be included in your responses where appropriate. For calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Pay attention to significant figures.

A student has 100. mL of 0.400 M $\text{CuSO}_4(\text{aq})$ and is asked to make 100. mL of 0.150 M $\text{CuSO}_4(\text{aq})$ for a spectrophotometry experiment. The following laboratory equipment is available for preparing the solution: centigram balance, weighing paper, funnel, 10 mL beaker, 150 mL beaker, 50 mL graduated cylinder, 100 mL volumetric flask, 50 mL buret, and distilled water.

- (a) Calculate the volume of 0.400 M $\text{CuSO}_4(\text{aq})$ required for the preparation.
- (b) Briefly describe the essential steps to most accurately prepare the 0.150 M $\text{CuSO}_4(\text{aq})$ from the 0.400 M $\text{CuSO}_4(\text{aq})$ using the equipment listed above.

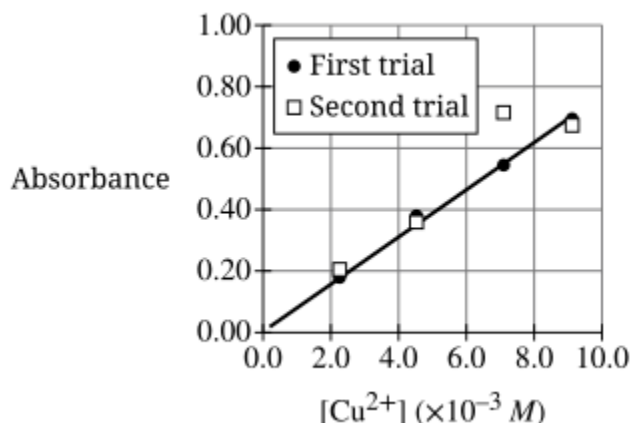
The student plans to conduct a spectrophotometric analysis to determine the concentration of $\text{Cu}^{2+}(\text{aq})$ in a solution. The solution has a small amount of $\text{Co}(\text{NO}_3)_2(\text{aq})$ present as a contaminant. The student is given the diagram below, which shows the absorbance curves for aqueous solutions of $\text{Co}^{2+}(\text{aq})$ and $\text{Cu}^{2+}(\text{aq})$.



- (c) The spectrophotometer available to the student has a wavelength range of 400 nm to 700 nm. What wavelength should the student use to minimize the interference from the presence of the $\text{Co}^{2+}(\text{aq})$ ions?

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9.



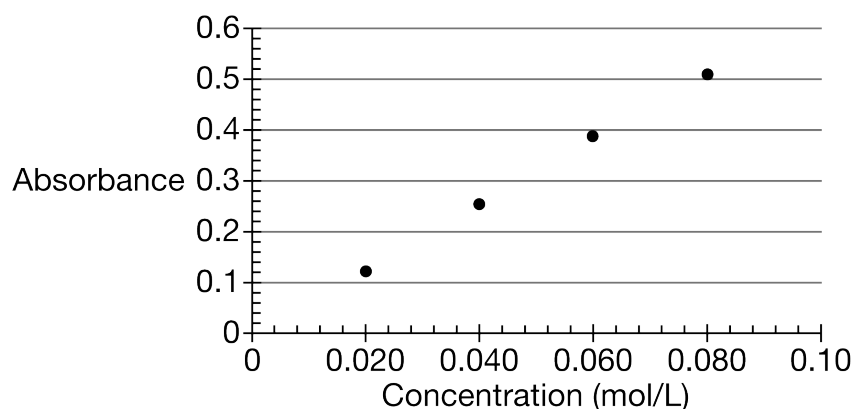
A student prepared four solutions of known $[\text{Cu}^{2+}]$ and measured the absorbance of each solution using the same cuvette. The graph shows the data for two absorbance measurements done for each solution. Which of the following identifies the most likely error that affected the absorbance recorded for the solution with $[\text{Cu}^{2+}] \approx 7 \times 10^{-3} \text{ M}$ in the second trial?

- (A) The cuvette was rinsed with water between measurements.
- (B) A fingerprint was left on the side of the cuvette facing the detector.
- (C) The absorbance was measured at a wavelength where Cu^{2+} has a lower molar absorptivity.
- (D) The cuvette was not filled with the same volume of solution.

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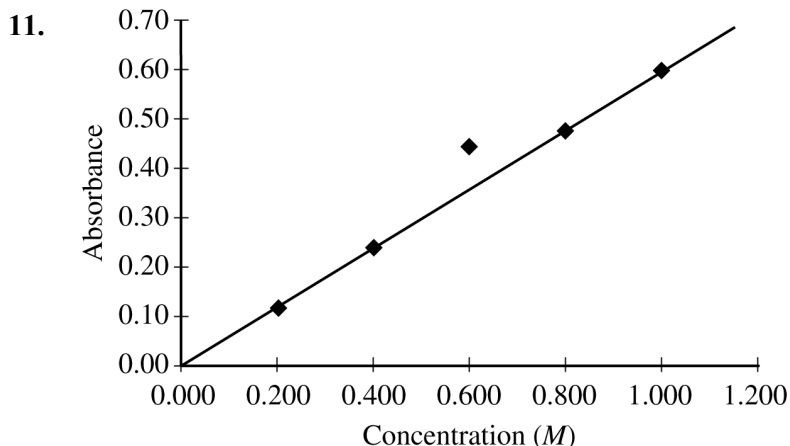
An experiment was performed to investigate the reaction between Zn metal and $\text{Ni}^{2+}(aq)$ at different concentrations. Because the $\text{Ni}^{2+}(aq)$ ion is green, the extent of the reaction was determined using spectrophotometric analysis. Four 20.0 mL standard solutions of $\text{Ni}^{2+}(aq)$ were prepared by dissolving $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ (molar mass 240 g/mol) in water. The absorbance of each solution was measured; the results are shown both in the table below and in the following plot of the absorbance data.

Solution	$[\text{Ni}^{2+}]$	Absorbance
1	0.020	0.12
2	0.040	0.25
3	0.060	0.39
4	0.080	0.51



10. What is the expected absorbance of a standard solution made by dissolving 0.0070 mol of $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ in water to make 100. mL of solution?
- (A) 0.0039
(B) 0.0080
(C) 0.045
(D) 0.45

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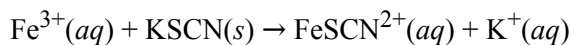


A student prepared five solutions of CuSO_4 with different concentrations, and then filled five cuvettes, each containing one of the solutions. The cuvettes were placed in a spectrophotometer set to the appropriate wavelength for maximum absorbance. The absorbance of each solution was measured and recorded. The student plotted absorbance versus concentration, as shown in the figure above. Which of the following is the most likely explanation for the variance of the data point for the 0.600 M CuSO_4 solution?

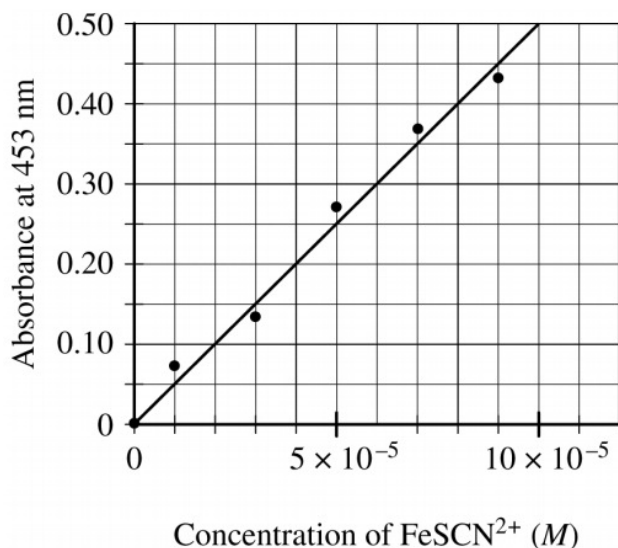
- (A) The cuvette into which the 0.600 M solution was placed had some water droplets inside.
- (B) The cuvette into which the 0.600 M solution was placed was filled slightly more than the other cuvettes.
- (C) The wavelength setting was accidentally moved away from that of maximum absorbance.
- (D) The cuvette used for the 0.600 M solution had not been wiped clean before being put in the spectrophotometer.

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12.

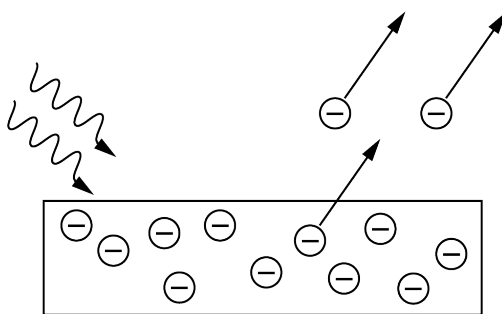


To determine the moles of $\text{Fe}^{3+}(\text{aq})$ in a 100. mL sample of an unknown solution, excess $\text{KSCN}(\text{s})$ is added to convert all the $\text{Fe}^{3+}(\text{aq})$ into the dark red species $\text{FeSCN}^{2+}(\text{aq})$, as represented by the equation above. The absorbance of $\text{FeSCN}^{2+}(\text{aq})$ at different concentrations is shown in the graph below.



If the absorbance of the mixture is 0.20 at 453 nm, how many moles of $\text{Fe}^{3+}(\text{aq})$ were present in the 100. mL sample? (Assume that any volume change due to adding the $\text{KSCN}(\text{s})$ is negligible.)

- (A) 4×10^{-4} mol
- (B) 3×10^{-4} mol
- (C) 4×10^{-6} mol
- (D) 3×10^{-6} mol



The diagram above represents the photoelectric effect for a metal. When the metal surface is exposed to light with increasing frequency and energy of photons, electrons first begin to be ejected from the metal when the energy of the photons is 3.3×10^{-19} J.

13. Which of the following is closest to the frequency of the light with photon energy of 3.3×10^{-19} J?

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- (A) $5.0 \times 10^{-53} \text{ s}^{-1}$
(B) $5.0 \times 10^{-16} \text{ s}^{-1}$
(C) $5.0 \times 10^{14} \text{ s}^{-1}$
(D) $5.0 \times 10^{52} \text{ s}^{-1}$

14.

Color	Wavelength
Red	647 – 760 nm
Orange	585 – 647 nm
Yellow	575 – 585 nm
Green	491 – 575 nm
Blue	424 – 491 nm
Violet	300 – 424 nm

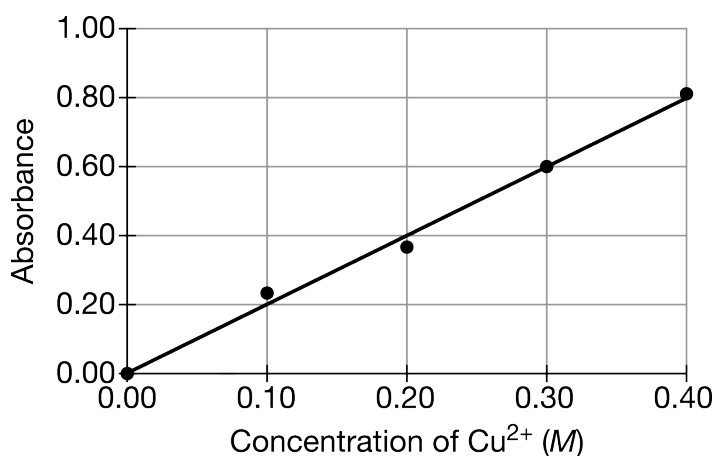
Using the wavelength information provided above, what is the color of the light?

- (A) Red
(B) Orange
(C) Yellow
(D) Blue
-

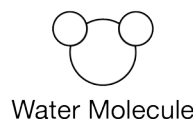
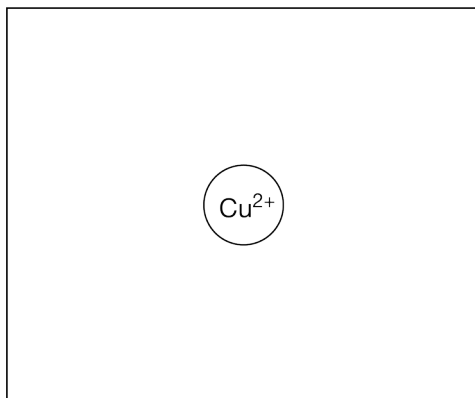
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15. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

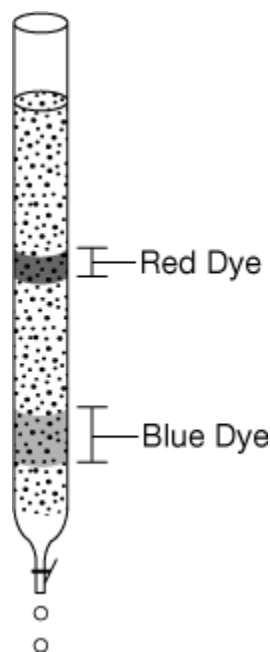
A student is given a sample of $\text{CuSO}_4(s)$ that contains a solid impurity that is soluble and colorless. The student wants to determine the amount of CuSO_4 in the sample and decides to use a spectrophotometer. First, the student prepares a calibration graph by measuring the absorbances of $\text{CuSO}_4(aq)$ solutions of known concentrations. The graph is shown below.



- (a) The student dissolves the entire impure sample of $\text{CuSO}_4(s)$ in enough distilled water to make 100. mL of solution. Then the student measures the absorbance of the solution and observes that it is 0.30. Determine the concentration of $\text{CuSO}_4(aq)$ in the solution.
- (b) Calculate the number of moles of CuSO_4 that were in the impure sample of $\text{CuSO}_4(s)$.
- (c) In addition to the number of moles of CuSO_4 calculated in part (b), what other quantity must be measured in order to calculate the mass percentage of CuSO_4 in the impure sample of $\text{CuSO}_4(s)$?
- (d) A particulate drawing of a single Cu^{2+} ion is shown in the following box. Draw two H_2O molecules in the box to show the most likely orientation of $\text{H}_2\text{O}(l)$ molecules around the Cu^{2+} ion.



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A student placed a sample of a food coloring that contains a mixture of a blue dye and a red dye at the top of a chromatography column filled with a nonpolar stationary phase. When water is poured through the column, two bands of colors are seen in the column, as shown in the diagram above.

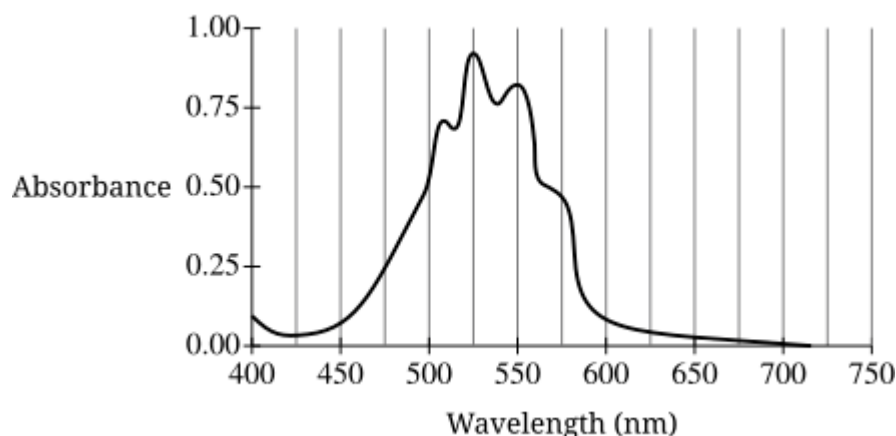
16. Which of the following techniques would be best to investigate molecular vibrations in molecules of the red dye?
- (A) Infrared spectroscopy
 - (B) UV / vis spectroscopy
 - (C) Mass spectrometry
 - (D) Gravimetric analysis
17. A student collects a sample of the blue dye from the column in a cuvette. Which of the following is least useful to determine the concentration of the sample using a spectrophotometer set at 640 nm?
- (A) The volume of the sample
 - (B) The path length of the cuvette
 - (C) The absorbance of the dye at 640 nm
 - (D) The molar absorptivity of the dye at 640 nm
-
18. Infrared spectroscopy is a useful tool for scientists who want to investigate the structure of certain molecules. Which of the following best explains what can occur as the result of a molecule absorbing a photon of infrared radiation?
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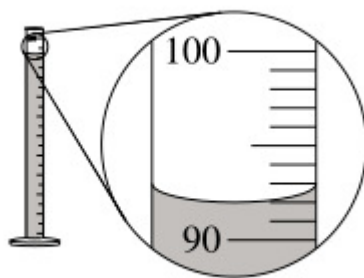
- (A) The energies of infrared photons are in the same range as the energies associated with changes between different electronic energy states in atoms and molecules. Molecules can absorb infrared photons of characteristic wavelengths, thus revealing the energies of electronic transitions within the molecules.
- (B) The energies of infrared photons are in the same range as the energies associated with different vibrational states of chemical bonds. Molecules can absorb infrared photons of characteristic wavelengths, thus revealing the types and strengths of different bonds in the molecules.
- (C) The energies of infrared photons are in the same range as the energies associated with different rotational states of molecules. Molecules can absorb infrared photons of characteristic wavelengths, thus revealing the energies of transition between different rotational energy states of the molecules.
- (D) The energies of infrared photons are in the same range as the total bond energies of bonds within molecules. Chemical bonds can be completely broken as they absorb infrared photons of characteristic wavelengths, thus revealing the energies of the bonds within the molecules.

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19. A student wants to determine the concentration of permanganate, $\text{MnO}_4^- (aq)$, in a solution. The student plans to use colorimetric analysis because solutions containing $\text{MnO}_4^- (aq)$ have a purple color.



- (a) To determine the optimum wavelength for an experiment that measures the concentration of $\text{MnO}_4^- (aq)$, the student takes a sample of the solution and measures the amount of light absorbed by the sample over a range of wavelengths. The data are plotted in the graph shown. Identify the optimum wavelength that the student should use for the experimental procedure.
- (b) The student uses a stock solution of $2.40 \times 10^{-3} \text{ M KMnO}_4 (aq)$ to prepare the standard solutions of $\text{MnO}_4^- (aq)$ that are needed to construct a calibration curve.



- (i) The student uses a 100.0 mL graduated cylinder to measure a certain volume of $\text{KMnO}_4 (aq)$ stock solution, as shown in the diagram given. What volume should the student record?
- (ii) Calculate the volume, in mL, of $2.40 \times 10^{-3} \text{ M KMnO}_4 (aq)$ that is required to produce 100.0 mL of a standard $1.68 \times 10^{-3} \text{ M MnO}_4^- (aq)$ solution.

The student designs the following procedure to produce a calibration curve.

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Step 1: Prepare several standard solutions that have known $\text{MnO}_4^- (aq)$ concentrations by dilution of the stock solution.

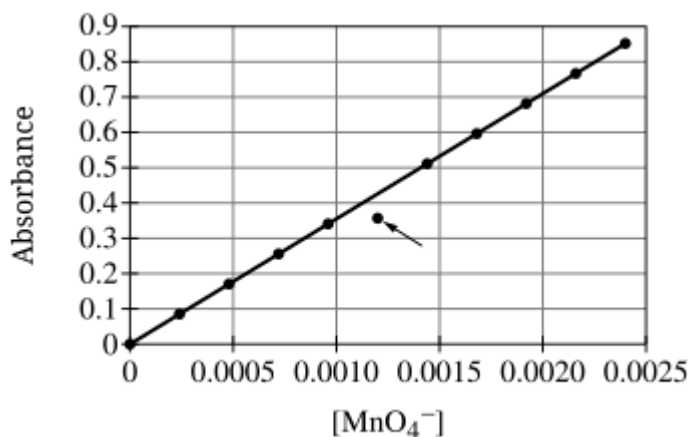
Step 2: Rinse the cuvette with distilled water.

Step 3: Rinse the cuvette with the standard solution and fill the cuvette with the standard solution.

Step 4: Measure the absorbance of the standard solution with the colorimeter.

Step 5: Repeat steps 2-4 for each of the standard solutions.

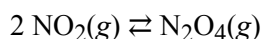
The data are plotted in the calibration curve shown. One of the data points (indicated with an arrow) on the calibration curve is below the line of best fit.



- (c) Assuming that all lab equipment is functioning properly, identify which one of the procedural steps the student could have executed incorrectly that would explain why the marked data point is below the line of best fit. Justify your answer.
20. Which statement correctly compares what occurs when molecules absorb photons in the microwave region with what occurs when molecules absorb photons in the infrared region?
- (A) Microwave photons cause the molecules to increase their rotational energy states, whereas infrared photons cause the molecules to increase their vibrational energy states.
- (B) Microwave photons cause electrons in the molecules to increase their electronic energy states, whereas infrared photons cause the molecules to increase their rotational energy states.
- (C) Microwave photons cause the molecules to increase their vibrational energy states, whereas infrared photons cause electrons in the molecules to increase their electronic energy states.
- (D) Microwave photons cause the molecules to increase their rotational energy states, whereas infrared photons cause electrons in the molecules to increase their electronic energy states.
21. N_2 molecules absorb ultraviolet light but not visible light. I_2 molecules absorb both visible and ultraviolet light. Which of the following statements explains the observations?

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- (A) More energy is required to make N_2 molecules vibrate than is required to make I_2 molecules vibrate.
- (B) More energy is required to remove an electron from an I_2 molecule than is required to remove an electron from a N_2 molecule.
- (C) Visible light does not produce transitions between electronic energy levels in the N_2 molecule but does produce transitions in the I_2 molecule.
- (D) The molecular mass of I_2 is greater than the molecular mass of N_2 .
- 22.** N_2 molecules absorb ultraviolet light but not visible light. I_2 molecules absorb both visible and ultraviolet light. Which of the following statements explains the observations?
- (A) More energy is required to make N_2 molecules vibrate than is required to make I_2 molecules vibrate.
- (B) More energy is required to remove an electron from an I_2 molecule than is required to remove an electron from a N_2 molecule.
- (C) Visible light does not produce transitions between electronic energy levels in the N_2 molecule but does produce transitions in the I_2 molecule.
- (D) The molecular mass of I_2 is greater than the molecular mass of N_2 .
-



dark brown colorless

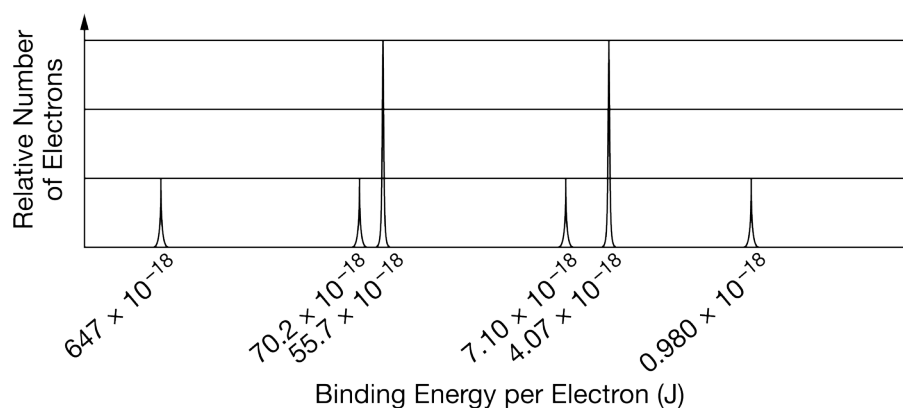
The dimerization of $\text{NO}_2(g)$, an exothermic process, is represented by the equation above.

- 23.** Which of the following experimental techniques will allow the most accurate determination of the concentration of $\text{NO}_2(g)$ at equilibrium?
- (A) Paper chromatography
- (B) Gravimetric analysis
- (C) Titration
- (D) Spectrophotometry
-

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24. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

The complete photoelectron spectrum of an element in its ground state is represented below.



- (a) Based on the spectrum,
- write the ground-state electron configuration of the element, and
 - identify the element.
- (b) Calculate the wavelength, in meters, of electromagnetic radiation needed to remove an electron from the valence shell of an atom of the element.

25.

Trial	$[\text{Cu}^{2+}]$	Absorbance
1	0.025	0.124
2	0.050	0.268
3	0.100	0.520
4	0.150	0.680

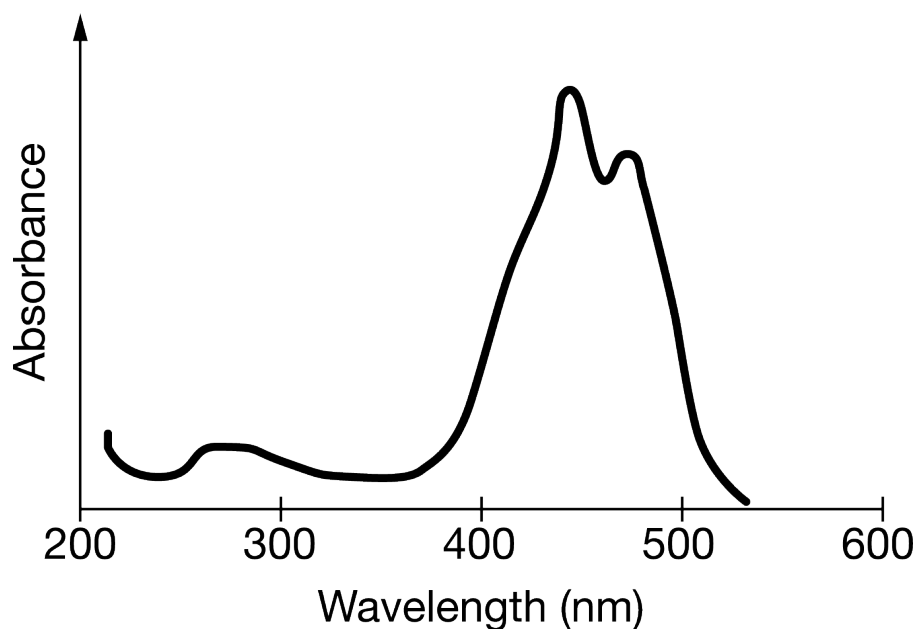
Using a spectrophotometer, a student measures the absorbance of four solutions of CuSO_4 at a given wavelength. The collected data is given in the table above. Which of the following is the most likely explanation for the discrepant data in trial 4?

- The solution was at a lower temperature than the solutions in the other trials.
- The measurement was made using a different spectrophotometer that uses a cell with a longer path length.
- The solution was saturated and the flow of light through the solution was restricted.
- The concentration of the solution was actually lower than 0.150 M .

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26. A student is given the task of making a calibration curve for $\text{CoCl}_2(aq)$ for use in a spectrophotometry experiment. The student pours a sample of $\text{CoCl}_2(aq)$ of known concentration into a cuvette with a 1.0 cm path length and tries to measure the absorbance at 500 nm, which is the wavelength of maximum absorbance. However, the absorbance is too high to give an accurate reading. Which of the following is the most appropriate next step for the student?
- (A) Switching to using a cuvette with a 1.5 cm path length
(B) Repeating the experiment after warming the solution to 40°C
(C) Diluting a measured volume of the solution to a new known volume using distilled water
(D) Pouring some solution out of the cuvette

27.



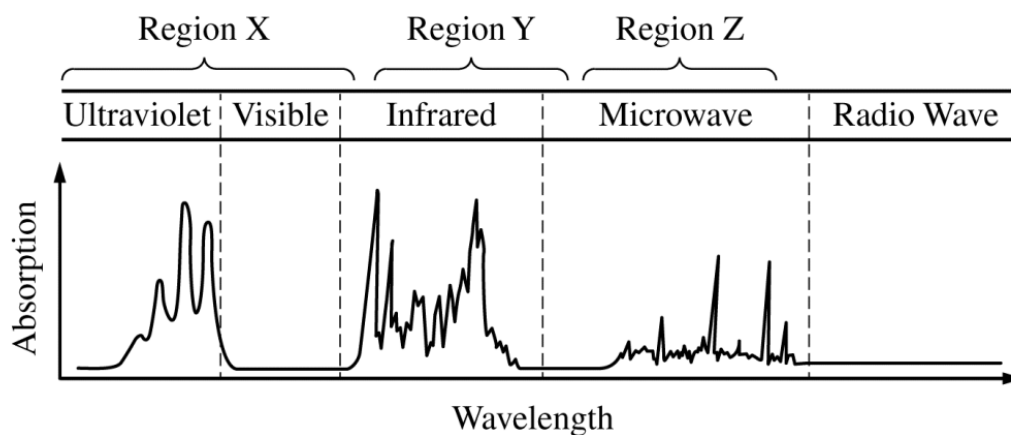
Beta-carotene is an organic compound with an orange color. The diagram above shows the ultraviolet spectrum of beta-carotene. Which of the following statements is true about the absorption bands in the spectrum?

- (A) The absorption band between 250 and 320 nm is due to transitions in electronic energy levels, and the absorption band between 380 and 520 nm is due to transitions in molecular vibrational levels.
- (B) The absorption band between 250 and 320 nm is due to transitions in molecular vibrational levels, and the absorption band between 380 and 520 nm is due to transitions in molecular rotational levels.
- (C) The two main absorption bands are associated with transitions in electronic energy levels. The band in the region corresponding to shorter wavelengths shows a lower absorbance than the band in the region corresponding to longer wavelengths.
- (D) The two main absorption bands are associated with transitions in molecular vibrational levels. The band in the region corresponding to shorter wavelengths shows a lower absorbance than the band in the region corresponding to longer wavelengths.

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28.

ABSORPTION SPECTRUM



The diagram above represents the absorption spectrum for a pure molecular substance. Which of the following correctly indicates the type of transition observed for the substance in each of the regions of the absorption spectrum?

	<u>Region X</u>	<u>Region Y</u>	<u>Region Z</u>
(A)	Molecular vibration	Molecular rotation	Electronic transition
(B)	<u>Region X</u>	<u>Region Y</u>	<u>Region Z</u>
	Electronic transition	Molecular rotation	Molecular vibration
(C)	<u>Region X</u>	<u>Region Y</u>	<u>Region Z</u>
	Molecular rotation	Molecular vibration	Electronic transition
(D)	<u>Region X</u>	<u>Region Y</u>	<u>Region Z</u>
	Electronic transition	Molecular vibration	Molecular rotation